

Notes: Edmond, Midrigan, and Xu (JPE, 2023)

How Costly Are Markups?

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Observed firm-level data and calibration targets	3
2.2	Key empirical objects in the benchmark calibration	4
2.3	Markup measurement and why cost weighting matters	4
2.4	Unobservables, firm heterogeneity, and timing	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Representative household and aggregate resource constraints	5
3.2	Firm technology, marginal cost, and markup pricing	6
3.3	Aggregation and the role of markup dispersion	6
3.4	Market structure variants	7
3.5	Planner's problem and the three channels of inefficiency	8
3.6	Policy implementation and decomposition	8
3.7	Calibration and welfare measurement	9
4	Identification and instruments (what makes the estimates credible)	9
4.1	What identifies the benchmark model	9
4.2	Why the paper does not feed in separately estimated markups	9
4.3	Why cost weighting is the right aggregation object	10
4.4	What makes the quantitative exercise credible	10
4.5	Main limitations	10
4.6	Relation to the literature	10
5	Tables and figures	11

5.1	Table 1 and Table 2: calibration and the markup-size relationship	11
5.2	Table 3 and Figure 1: markup dispersion and productivity losses	11
5.3	Figure 2: equilibrium versus planner allocations	12
5.4	Table 4 and Figure 3: total welfare costs and policy decomposition	13
5.5	Figure 4 and Figure 5: why subsidizing entry does little	13
5.6	Table 5, Table 6, and Table 7: oligopoly results	14
5.7	Appendix Figure A1 and Appendix Figure B1: how to read markup time series . . .	15
6	Short summary	15
7	Conclusion	16
7.1	Core conclusion (what the paper establishes)	16
7.2	Economic intuition	16
7.3	Takeaway	17

1 Big picture

- **Question.** How large are the welfare costs of markups once we take seriously a dynamic economy with heterogeneous firms, endogenous entry, and endogenous markup dispersion?
- **Setting.** The paper builds a dynamic macro-IO model in which firms differ in productivity, enter by paying a sunk cost, exit randomly, and charge markups that vary with firm size. The model is calibrated to U.S. Census of Manufactures data from 1972 to 2012.
- **Core challenge.** Markups matter at both the aggregate and cross-sectional levels. The aggregate level of markups depresses employment and investment like a uniform output tax, but markup dispersion also reallocates production across firms. A representative-firm model misses this second margin.
- **Main theoretical result.** The relevant aggregate wedge is the *cost-weighted average markup*, not the usual sales-weighted arithmetic average. Markups reduce welfare through three channels: an aggregate-markup wedge, misallocation from markup dispersion, and a distorted entry margin.
- **Why this matters.** The paper changes how one should read empirical markup data. A rise in the sales-weighted average markup may partly reflect greater dispersion rather than a larger aggregate wedge in firms' static and dynamic first-order conditions.
- **Main empirical strategy.** The authors calibrate the model to match sales concentration, materials shares, and the empirical relationship between markups and market shares (or, in the oligopoly version, between sector labor shares and concentration). They do *not* take separately estimated firm-level markups as primitive inputs.
- **Main quantitative result.** Welfare costs are large and strongly convex in the target aggregate markup M . In the benchmark Kimball-demand calibration, eliminating all markup distortions yields consumption-equivalent gains of about **1.34%** when $M = 1.05$, **8.67%** when $M = 1.15$, **23.64%** when $M = 1.25$, and **49.66%** when $M = 1.35$.
- **Main decomposition result.** Across specifications, the **aggregate-markup** and **misallocation** channels account for almost all the welfare costs. The **entry channel** is much smaller.
- **Policy result.** A uniform output subsidy that removes only the aggregate-markup wedge captures a large share of the gains. Size-dependent subsidies that eliminate misallocation matter too, especially when markup dispersion is high. By contrast, subsidizing entry is a weak instrument because more entry barely changes the aggregate markup.
- **Economic takeaway.** The paper's message is not just "markups are costly." It is that the welfare cost of markups depends on the level of the aggregate wedge, the amount of markup dispersion, and how both interact with firm dynamics and entry.

2 Data and measurement (key objects)

2.1 Observed firm-level data and calibration targets

- **Primary data.** The main empirical source is the *U.S. Census of Manufactures*, 1972–2012.

- **Unit of observation.** The core moments are constructed using firm-level revenue, labor expenditure, and sales concentration within **6-digit NAICS** manufacturing sectors.
- **What is observed and what is not.** The Census gives firm revenue data but not firm-level prices and quantities separately. This is crucial because it means markup levels cannot be backed out without additional structure.
- **Why these data are useful.** The paper needs moments that discipline how markups covary with firm size and how concentrated production is. Census data are well suited for both.

2.2 Key empirical objects in the benchmark calibration

- **Target top-5% sales share:** 0.57.
- **Target materials share:** 0.45.
- **Target slope coefficient in the markup-size relationship:** $\hat{\beta} = 0.16$.
- **Target aggregate markup range:** roughly $M \in [1.1, 1.4]$ in Census-based calculations, so the paper reports results for multiple target values of M rather than pretending one precise number is known.

Interpretation. The benchmark exercise is not “estimate one true markup and run with it.” Instead, it asks: for empirically plausible values of the aggregate markup, what welfare costs and policy implications does the model deliver?

2.3 Markup measurement and why cost weighting matters

The paper’s key measurement object is the *aggregate markup* M . At the firm level, with markup pricing,

$$p_{it}(s) = m_{it}(s) \frac{\Omega_t}{z_i(s)},$$

so markups vary with the demand elasticity faced by the firm.

At the sector level,

$$m_t(s) = \int m_{it}(s) \frac{l_{it}(s)}{l_t(s)} di,$$

and at the aggregate level,

$$M_t = \int_0^1 m_t(s) \frac{l_t(s)}{\tilde{L}_t} ds.$$

- This says the relevant aggregate markup is an **employment- or cost-weighted arithmetic average** of lower-level markups.
- Equivalently, it is a **sales-weighted harmonic average**, not a sales-weighted arithmetic average.
- The paper then shows that if M^f denotes the usual sales-weighted arithmetic average markup, then

$$M^f = M + \frac{1}{M} \text{Var}[m_i].$$

Interpretation. The sales-weighted arithmetic average markup mixes together two things:

- the true aggregate wedge M , and
- a dispersion term that reflects misallocation.

So if one uses the sales-weighted arithmetic average as “the aggregate markup,” one overstates the wedge relevant for aggregate employment and investment decisions.

2.4 Unobservables, firm heterogeneity, and timing

- **Firm heterogeneity.** Firms draw a one-time productivity (or, in one extension, quality) draw upon entry.
- **Entry.** Firms enter by paying a sunk labor cost.
- **Exit.** Exit is random with annual probability $J = 0.04$.
- **Dynamics.** There is no aggregate uncertainty; the dynamic problem is about transition paths in capital and the mass of firms after policy changes.
- **Unobservables.** Because prices and quantities are not separately observed in the data, the model must provide the mapping from labor shares and market shares to markups.

3 Models and empirical specifications (all equations + interpretation)

3.1 Representative household and aggregate resource constraints

The representative consumer solves

$$\sum_{t=0}^{\infty} \beta^t \left(\log C_t - \omega \frac{L_t^{1+\eta}}{1+\eta} \right)$$

subject to

$$C_t + I_t = W_t L_t + R_t K_t + \Pi_t,$$

with investment law of motion

$$I_t = K_{t+1} - (1 - \delta)K_t.$$

Final output is used for consumption, investment, and materials:

$$C_t + I_t + X_t = Y_t.$$

Across sectors, gross output is aggregated through

$$Y_t = \left(\int_0^1 y_t(s)^{\frac{h-1}{h}} ds \right)^{\frac{h}{h-1}}.$$

Interpretation. The household side is standard, but the important modeling choice is that the final good is also used as *materials*. This matters because the aggregate markup distorts not only labor and capital demand, but also the value-added versus materials margin.

3.2 Firm technology, marginal cost, and markup pricing

A firm in sector s with productivity $z_i(s)$ produces gross output

$$y_{it}(s) = z_i(s) \left[f^{1/\nu} v_{it}(s)^{\frac{\nu-1}{\nu}} + (1-f)^{1/\nu} x_{it}(s)^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}},$$

where value added is

$$v_{it}(s) = k_{it}(s)^\alpha l_{it}(s)^{1-\alpha}.$$

Cost minimization implies marginal cost

$$\text{MC}_{it}(s) = \frac{\Omega_t}{z_i(s)}.$$

Firms then choose prices as a markup over marginal cost:

$$p_{it}(s) = m_{it}(s) \frac{\Omega_t}{z_i(s)}, \quad m_{it}(s) = \frac{\iota_{it}(s)}{\iota_{it}(s) - 1},$$

where $\iota_{it}(s)$ is the demand elasticity the firm faces.

Profits are

$$\pi_{it}(s) = \left(1 - \frac{1}{m_{it}(s)} \right) p_{it}(s) y_{it}(s).$$

Labor shares satisfy

$$\frac{W_t l_{it}(s)}{p_{it}(s) y_{it}(s)} = \frac{(1-\alpha) z_t}{m_{it}(s)}.$$

Interpretation. This labor-share equation is the bridge from data to markups. Cross-sectional variation in labor shares within the model is driven by cross-sectional variation in markups, which is why the markup-size relationship is empirically informative.

3.3 Aggregation and the role of markup dispersion

Sectoral gross output can be written as

$$y_t(s) = z_t(s) F(k_t(s), l_t(s), x_t(s)),$$

where sector productivity is the firm-size-weighted harmonic average of firm productivity:

$$z_t(s) = \left(\int \frac{q_{it}(s)}{z_i(s)} di \right)^{-1}.$$

Aggregate gross output is

$$Y_t = Z_t F(K_t, \tilde{L}_t, X_t),$$

with aggregate productivity

$$Z_t = \left(\int_0^1 \frac{q_t(s)}{z_t(s)} ds \right)^{-1}.$$

Aggregate markup satisfies

$$1 = \frac{M_t \Omega_t}{Z_t}.$$

The paper then shows that markup dispersion affects productivity through the induced size distribution of firms and sectors. In particular,

$$Z_t = \left(\int_0^1 \left(\frac{m_t(s)}{M_t} \right)^{-h} z_t(s)^{h-1} ds \right)^{\frac{1}{h-1}}.$$

Interpretation.

- If markups were the same across firms, there would be no markup-driven misallocation.
- When larger firms face less elastic demand and charge higher markups, they become too small relative to the planner's allocation.
- Less productive, low-markup firms become too large.

3.4 Market structure variants

The paper studies several within-sector environments.

Benchmark: Kimball demand. In the benchmark model with monopolistic competition and Kimball demand,

$$\Upsilon'(q) = \frac{\bar{j} - 1}{\bar{j}} \exp\left(\frac{1 - q^{\epsilon/\bar{j}}}{\epsilon}\right), \quad j(q) = \bar{j} q^{-\epsilon/\bar{j}}.$$

Interpretation. Larger firms face less elastic demand when $\epsilon > 0$, so they charge higher markups. The superelasticity $\epsilon/\bar{j}$ is exactly the parameter that links market shares to markups in the benchmark case.

Quality-heterogeneity extension. If heterogeneity is in quality rather than productivity, the paper shows the markup-size relationship becomes

$$\frac{1}{m_t(z)} + \log\left(1 - \frac{1}{m_t(z)}\right) = a + b \log q_t(z) - b \log z,$$

so market share is no longer a sufficient statistic because unobserved quality also matters.

Translog extension. With symmetric translog demand,

$$m + \log m = 1 + \log\left(\frac{z}{z_t^*}\right), \quad m_t(z) = 1 + \frac{1}{\bar{j}} q_t(z).$$

The paper derives closed-form expressions for the productivity cutoff z_t^* and the aggregate markup M_t .

Oligopoly extension. With oligopolistic competition and nested CES demand, within-sector concentration matters directly and the model generates both within- and between-sector variation in markups.

Interpretation. The aggregation results are robust across market structures, but the quantitative importance of markup dispersion and the entry margin is not. Oligopoly tends to generate more dispersion and larger welfare costs.

3.5 Planner's problem and the three channels of inefficiency

The planner chooses how many varieties to create and how to allocate factors so as to maximize representative-household welfare. Two planner conditions summarize the economics.

First, the planner's intratemporal conditions imply no aggregate wedge between private and social marginal products. Compared with the decentralized economy, the aggregate markup acts like a **uniform output tax** reducing labor, capital, and materials.

Second, the planner's entry condition is

$$kW_t^* = \beta \sum_{j=1}^{\infty} [\beta(1-J)]^{j-1} \frac{C_t^*}{C_{t+j}^*} \frac{dZ_{t+j}^*}{dN_{t+j}^*} \frac{Y_{t+j}^*}{Z_{t+j}^*} N_{t+j}^*.$$

Interpretation. Comparing the planner's entry condition with free entry in the market reveals three channels through which markups distort welfare:

- **Aggregate-markup channel:** M_t acts like a common tax wedge.
- **Misallocation channel:** cross-firm markup variation distorts the allocation of labor, capital, and materials.
- **Entry channel:** private incentives to create new varieties differ from social incentives.

3.6 Policy implementation and decomposition

The efficient allocation can be implemented with a direct size-dependent subsidy schedule

$$T_t(q) = (\Upsilon(q) - \Upsilon'(q)q) D_t Y_t.$$

This induces marginal-cost pricing:

$$p_t(z) = \frac{\Omega_t}{z},$$

so firm-level and aggregate wedges both become 1.

The paper then generalizes the policy to

$$T_t(q) = (a_0 \Upsilon(q) + a_1 \Upsilon'(q)q) D_t Y_t.$$

Three cases matter:

- **Efficient allocation:** $a_0 = 1, a_1 = -1$.
- **Uniform subsidy:** $a_0 = 0, a_1 = x > 0$. This removes only the aggregate-markup wedge.
- **Size-dependent subsidy:** $a_0 = 1/(1+x), a_1 = -1$. This removes markup dispersion while leaving the aggregate wedge in place.

Interpretation. This decomposition is what lets the paper say quantitatively how much of the welfare loss comes from the level of markups, how much from dispersion, and how much from distorted entry.

3.7 Calibration and welfare measurement

- Assigned parameters include $\beta = 0.96$, depreciation $\delta = 0.06$, exit rate $J = 0.04$, capital share $\alpha = 1/3$, Frisch elasticity $1/\eta = 1$, and elasticity of substitution between value added and materials $\nu = 0.5$.
- The benchmark calibration jointly chooses the Pareto tail y , average demand elasticity $\bar{j}$, superelasticity $\epsilon/\bar{j}$, and value-added weight f to match the empirical targets.
- Welfare is measured as the consumption-equivalent gain from moving from the distorted steady state to the efficient steady state, taking the full transition path into account.

Interpretation. Transitional dynamics matter because removing markup distortions requires a large buildup in capital and sometimes in the mass of firms. A static comparison would overstate the immediate welfare gain.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the benchmark model

This paper is a calibration-and-discipline exercise rather than a conventional IV design. The key identification logic is:

- **Sales concentration** identifies the Pareto tail parameter y .
- **The markup-market share slope** identifies the superelasticity $\epsilon/\bar{j}$ in the benchmark Kimball model.
- **The target aggregate markup** M identifies the level of demand elasticity $\bar{j}$.
- **The materials share** identifies the weight on value added f .

In the oligopoly version:

- CR4 and CR20 pin down productivity dispersion and entry costs,
- the target aggregate markup helps pin down the within-sector elasticity g ,
- the slope of sector labor shares on sector HHIs helps pin down the gap between the within- and between-sector elasticities $(g - h)$.

4.2 Why the paper does not feed in separately estimated markups

- The data do not separately observe firm prices and quantities.
- With revenue data alone, markup levels are hard to disentangle from output elasticities in production.
- Even if one estimated firm-level markups from revenue data, those estimates could confound market power with other wedges such as taxes, factor-adjustment costs, or price rigidities.

Interpretation. The paper’s strategy is deliberately structural: use the model-implied relationship between size and markups to recover a coherent markup distribution, rather than taking raw markup estimates as literal truth.

4.3 Why cost weighting is the right aggregation object

- The representative household’s and firms’ first-order conditions depend on the wedge between the aggregate price level and aggregate marginal cost.
- That wedge is the cost-weighted average markup M , not the sales-weighted arithmetic average M^f .
- The decomposition

$$M^f = M + \frac{1}{M} \text{Var}[m_i]$$

means that a rise in M^f can come from either a larger aggregate wedge or more dispersion.

Interpretation. This is one of the paper’s most useful lessons for empirical macro: a rise in the sales-weighted average markup does not automatically imply the same-sized rise in the aggregate wedge relevant for labor and investment.

4.4 What makes the quantitative exercise credible

- The benchmark Kimball model is flexible enough to fit both the concentration target and the observed positive slope between markups and market shares.
- The paper reports results for a **range** of aggregate markup targets rather than anchoring everything to one controversial number.
- The authors show robustness across different market structures: Kimball monopolistic competition, quality heterogeneity, translog demand, and oligopoly.

4.5 Main limitations

- The paper attributes the systematic relationship between labor shares and firm size to market power; if some of that relationship reflects overhead costs or other wedges, the misallocation losses are an upper bound.
- The model abstracts from other market-specific sources of markup variation, such as location-specific competition or product-level heterogeneity within the same firm.
- There are no positive innovation or knowledge spillovers from markup-funded entry; if such spillovers matter, the welfare cost of markups may be overstated.

4.6 Relation to the literature

- Relative to the recent empirical markup literature, the paper’s main contribution is aggregation: how micro markup heterogeneity maps into macro wedges.

- Relative to misallocation papers such as Hsieh-Klenow, the paper endogenizes markup dispersion through market structure rather than taking wedges as exogenous.
- Relative to Bilbiie-Ghironi-Melitz type representative-firm models, it shows that firm heterogeneity is essential for understanding why entry subsidies do so little to the aggregate markup.

5 Tables and figures

5.1 Table 1 and Table 2: calibration and the markup-size relationship

TABLE 1
PARAMETERIZATION

A. ASSIGNED PARAMETERS					
Parameter	Value				
Discount factor, β	.96				
Depreciation rate, δ	.06				
Exit rate, φ	.04				
Elasticity of value-added to capital, α	1/3				
Elasticity of labor supply, ν	1				
Elasticity of substitution between value-added and materials, θ	.5				
B. TARGETS AND PARAMETERS					
	Data	Cases			
Calibration targets:					
Aggregate markup, $\mathcal{M}$	1.1~1.4	1.05	1.15	1.25	1.35
Top 5% sales share	.57	.57	.57	.57	.57
Materials share	.45	.45	.45	.45	.45
Regression coefficient, $\hat{b}$	.16	.16	.16	.16	.16
Parameters:					
Pareto tail, ξ		20.70	6.84	4.07	2.89
Demand elasticity, $\bar{\sigma}$		29.10	10.86	7.21	5.66
Superelasticity, $\varepsilon/\bar{\sigma}$		.16	.16	.16	.16
Weight on value-added, ϕ		.51	.43	.33	.21

NOTE.—Panel A reports assigned parameters held constant through all our quantitative exercises. Panel B reports calibrated parameters for our benchmark model with monopolistic competition and Kimball demand. We report four cases corresponding to alternative targets for the level of the aggregate markup ($\mathcal{M} = 1.05, 1.15, 1.25$ and 1.35) over the range of $\mathcal{M}$ implied by the US Census of Manufactures from 1972 to 2012, as discussed in app. B. For each $\mathcal{M}$ we calibrate the Pareto tail ξ , demand elasticity $\bar{\sigma}$, superelasticity $\varepsilon/\bar{\sigma}$, and weight on value-added ϕ to match the targets shown in panel B. For each model we choose the superelasticity $\varepsilon/\bar{\sigma}$ so that the slope coefficient b from eq. (59) in the model matches the estimated slope coefficient $\hat{b}$. See the text for more details.

TABLE 2
RELATIONSHIP BETWEEN MARKUPS AND MARKET SHARES

	Dependent Variable: $1/\mu_i(s) + \log(1 - (1/\mu_i(s)))$		
$\log \omega_i(s)$	.063 (.001)	.162 (.002)	.187 (.003)
Sector $\times$ year fixed effects	Y	Y	Y
Firm fixed effects		Y	Y
Firm age			Y
R^2	.084	.531	.540
Observations	609,000	369,000	315,000

NOTE.—Firm-level markups $\mu_i(s)$ are constructed from the US Census of Manufactures from 1972 to 2012, as discussed in the text. Market shares $\omega_i(s)$ of firm i are constructed within each 6-digit NAICS sector s . We include sector $\times$ year fixed effects to control for sector-specific shifts in the Kimball demand index $d_i(s)$. Our benchmark specification also includes firm fixed effects to control for any time-invariant firm-specific component of demand. Results are robust to including firm age. Standard errors (in parentheses) are clustered at the firm level.

Read:

- Table 1 reports the assigned macro parameters and the calibrated structural parameters for different targets of the aggregate markup M .
- As M rises from 1.05 to 1.35, the calibration requires a much lower average demand elasticity and a much fatter-tailed productivity distribution. For example, the Pareto tail falls from about 20.70 to 2.89, and the average demand elasticity falls from about 29.10 to 5.66.
- Table 2 is the empirical backbone of the calibration. The preferred specification, with sector-year and firm fixed effects, gives a slope of about **0.162** in the transformed regression of markups on log market shares.
- The message is that larger firms systematically have higher markups, and the model is disciplined to replicate exactly that fact.

5.2 Table 3 and Figure 1: markup dispersion and productivity losses

Read:

	AGGREGATE MARKUP, $\mathcal{M}$			
	1.05	1.15	1.25	1.35
Cost-weighted distribution of markups:				
25th percentile markup	1.04	1.11	1.17	1.23
50th percentile markup	1.05	1.14	1.23	1.31
75th percentile markup	1.06	1.18	1.31	1.43
90th percentile markup	1.07	1.23	1.40	1.58
99th percentile markup	1.11	1.35	1.63	1.97
Aggregate-productivity losses (%):				
Gross output	.28	.97	1.83	2.86
Value-added	.61	2.71	6.08	10.73
Value-added, $\mathcal{M} = 1$	.51	1.85	3.63	5.85

NOTE.—Cost-weighted steady-state distribution of markups and aggregate-productivity losses for four calibrations of our benchmark model, corresponding to targets for the aggregate markup $\mathcal{M} = 1.05, 1.15, 1.25$ and 1.35 . Gross-output aggregate-productivity loss is $(Z - Z^*)/Z^* \times 100$, and similarly for the value-added aggregate-productivity loss. To isolate the effect of misallocation on value-added aggregate productivity, we also report the value-added aggregate-productivity loss with the same amount of markup dispersion but holding $\mathcal{M} = 1$, to eliminate the distortion between value-added and materials; see text for details.

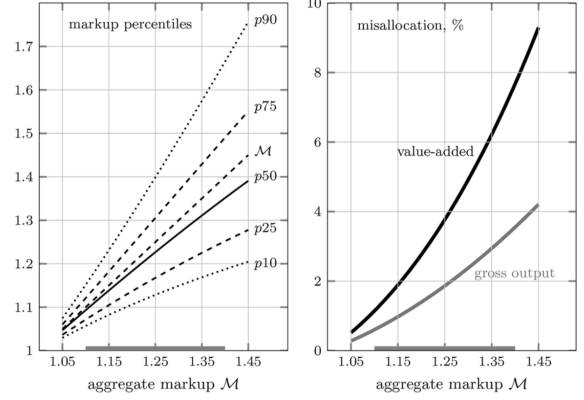


FIG. 1.—Markup distribution and misallocation in benchmark model. *Left*, cost-weighted steady-state markup distribution in our benchmark model with monopolistic competition and Kimball demand for a range of targets for the aggregate markup $\mathcal{M}$. For each $\mathcal{M}$ we recalibrate the Pareto tail ξ , demand elasticity $\bar{\sigma}$, superelasticity $\varepsilon/\bar{\sigma}$, and weight on value-added ϕ to match the calibration targets in table 1. *Right*, implied amounts of misallocation in aggregate gross output and aggregate value-added. To isolate the effect of misallocation on value-added aggregate productivity, we report the value-added aggregate productivity loss with the same amount of markup dispersion but holding $\mathcal{M} = 1$ to eliminate the distortion between value-added and materials; see text for details. Shaded interval indicates the range of $\mathcal{M}$ implied by the US Census of Manufactures from 1972 to 2012, as discussed in appendix B.

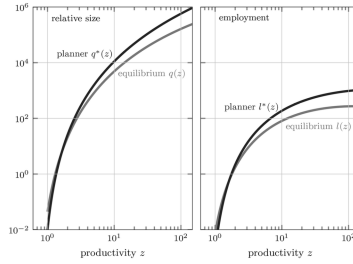


FIG. 2.—Equilibrium and planner allocations. *Left*, the equilibrium relative size $q(z)$ and the planner's relative size $q^*(z)$ as functions of productivity for our benchmark economy with $\mathcal{M} = 1.15$. *Right*, the equilibrium employment $l(z)$ and the planner's employment $l^*(z)$ for the same economy. More productive firms have higher markups, produce too little, and employ too little, compared to the planner's allocation. Less productive firms produce too much and employ too much, compared to the planner's allocation. In this figure, aggregate employment in the decentralized equilibrium is the same as aggregate employment for the planner. Our measure of misallocation is the aggregate-output loss implied by the equilibrium allocation relative to the planner's allocation.

- Table 3 shows that markup dispersion rises sharply as the targeted aggregate markup rises. In the benchmark model, the 99th percentile markup goes from about **1.11** when $M = 1.05$ to about **1.97** when $M = 1.35$.
- Gross-output productivity losses from misallocation rise from about **0.28%** to **2.86%**.
- Value-added productivity losses rise more sharply, from about **0.61%** to **10.73%**, because the aggregate markup also distorts the materials versus value-added margin.
- The “value-added, $M = 1$ ” row is important: it isolates the effect of markup *dispersion* holding the aggregate level of markups fixed.
- Figure 1 visualizes the same point: as M rises, the markup distribution stretches out and productivity losses become much larger.

5.3 Figure 2: equilibrium versus planner allocations

Read:

- Figure 2 is the clearest picture of misallocation in the paper.

- High-productivity firms are too small in the decentralized equilibrium relative to the planner; they produce and employ too little.
- Low-productivity firms are too large.
- This is exactly what one would expect when high-productivity firms face less elastic demand, charge higher markups, and therefore contract too much relative to the social optimum.

5.4 Table 4 and Figure 3: total welfare costs and policy decomposition

	STEADY-STATE COMPARISONS (% change)						WELFARE (% change)
	Y	C	L	N	K	Z	
$\mathcal{M} = 1.05$:							
Efficient	15.3	11.0	6.0	12.3	24.1	.9	1.34
Uniform subsidy	13.7	9.1	5.7	3.4	22.0	.2	.65
Size-dependent subsidy	1.4	1.7	.3	8.1	1.7	.7	.71
Entry subsidy	1.1	1.3	.4	11.3	1.4	.6	.06
$\mathcal{M} = 1.15$:							
Efficient	59.6	44.5	18.0	20.1	100.4	4.1	8.67
Uniform subsidy	51.8	35.8	17.0	9.5	88.5	1.5	5.90
Size-dependent subsidy	5.3	6.2	1.0	8.3	6.6	2.3	2.87
Entry subsidy	6.3	7.4	2.4	20.0	8.1	3.0	.56
$\mathcal{M} = 1.25$:							
Efficient	134.0	102.3	30.1	26.7	246.2	8.9	23.64
Uniform subsidy	112.7	79.7	28.2	15.0	208.2	3.9	17.36
Size-dependent subsidy	10.8	12.5	1.8	8.1	13.6	4.1	6.26
Entry subsidy	17.4	20.3	6.0	29.1	23.0	7.4	1.98
$\mathcal{M} = 1.35$:							
Efficient	263.2	203.4	42.1	32.2	540.3	15.1	49.66
Uniform subsidy	213.5	152.9	39.2	19.8	435.1	7.4	37.41
Size-dependent subsidy	18.4	20.9	2.7	7.6	23.6	6.0	11.32
Entry subsidy	38.6	44.9	11.9	39.0	52.6	14.0	5.11

NOTE.—The first six columns report the percentage change from the initial distorted steady state to the new steady state. The last column reports the consumption-equivalent welfare gains (including transitional dynamics). For each $\mathcal{M}$ we recalibrate the Pareto tail ξ , demand elasticity $\bar{\sigma}$, superelasticity $\varepsilon/\bar{\sigma}$, and weight on value-added ϕ . The alternative policies are (i): the efficient allocation, where all markups are removed; (ii) a uniform subsidy that eliminates the aggregate markup; (iii) size-dependent subsidies that eliminate misallocation and the entry distortion; and (iv) the uniform entry subsidy that leads to the largest welfare gain.

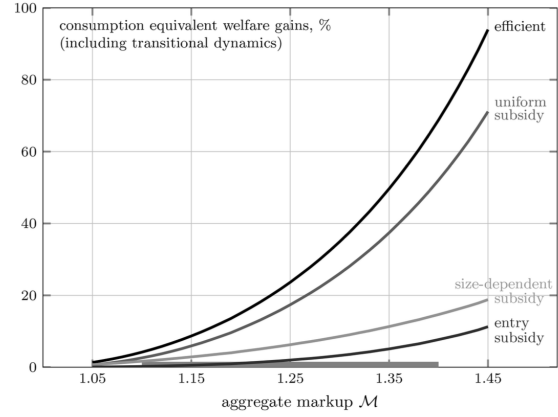


FIG. 3.—Welfare gains from alternative policies: consumption-equivalent welfare gains (including transitional dynamics), from the initial distorted steady state to the new steady state for a range of targets for the aggregate markup $\mathcal{M}$. For each $\mathcal{M}$ we recalibrate the Pareto tail ξ , demand elasticity $\bar{\sigma}$, superelasticity $\varepsilon/\bar{\sigma}$, and weight on value-added ϕ . The alternative policies are (i) the efficient allocation, where all markups are removed; (ii) a uniform subsidy that eliminates the aggregate markup; (iii) size-dependent subsidies that eliminate misallocation and the entry distortion; and (iv) the uniform entry subsidy that leads to the largest welfare gain. The shaded interval indicates the range of $\mathcal{M}$ implied by the US Census of Manufactures from 1972 to 2012, as discussed in appendix B.

Read:

- Table 4 is the paper’s main quantitative table for the benchmark Kimball model.
- The efficient allocation yields consumption-equivalent gains of **1.34%**, **8.67%**, **23.64%**, and **49.66%** as M rises from 1.05 to 1.35.
- The uniform subsidy that only removes the aggregate-markup wedge captures a large share of those gains: for example, **5.90%** when $M = 1.15$ and **37.41%** when $M = 1.35$.
- Size-dependent subsidies that remove misallocation and partly correct entry generate smaller but still meaningful gains, especially when M is large: **0.71%** at $M = 1.05$ and **11.32%** at $M = 1.35$.
- The optimal entry subsidy is tiny by comparison: **0.06%**, **0.56%**, **1.98%**, and **5.11%** across the same four values of M .
- Figure 3 shows that all these welfare curves are increasing in M , but the efficient-allocation curve is strongly convex. That convexity is why high aggregate markups imply very large welfare losses.

5.5 Figure 4 and Figure 5: why subsidizing entry does little

Read:

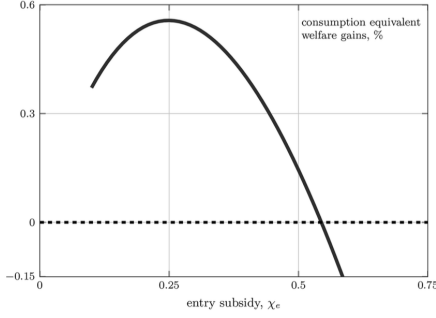


FIG. 4.—Optimal entry subsidy, $\mathcal{M} = 1.15$: consumption-equivalent welfare gains (including transitional dynamics) as a function of the entry subsidy χ_e for aggregate markup $\mathcal{M} = 1.15$. The welfare gains for entry subsidies reported in table 4 are for the optimal entry subsidies, that is, for the peak of such curves for each $\mathcal{M}$. In our benchmark calibration there is insufficient entry in the initial distorted steady state, so the optimal entry subsidy is positive. But entry subsidies that are too large lead to welfare losses.

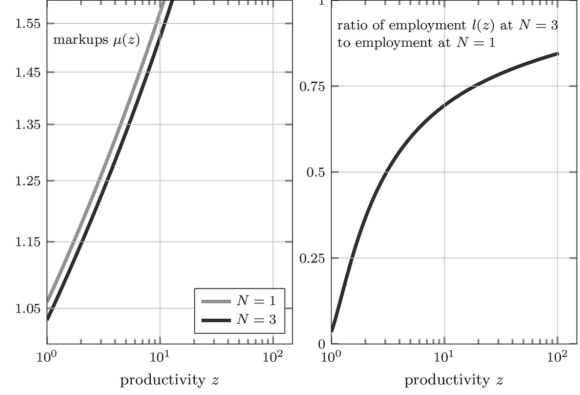


FIG. 5.—Effect of entry subsidy on markups, $\mathcal{M} = 1.15$. *Left*, steady-state markups $\mu(z)$ for an economy with mass of firms $N = 1$ and an entry subsidy chosen to triple the mass of firms to $N = 3$. *Right*, ratio of employment $l(z)$ at $N = 3$ to that at $N = 1$. Small, low-markup firms contract by more than large, high-markup firms, so that high-markup firms get relatively more weight in the aggregate-markup calculation. Because of this, the aggregate markup hardly changes. In this example, the aggregate markup barely changes, from $\mathcal{M} = 1.150$ to $\mathcal{M} = 1.146$, even though the mass of firms triples.

- Figure 4 plots welfare gains as a function of the entry subsidy. The curve is hump-shaped: some entry subsidy is optimal in the benchmark, but large subsidies reduce welfare.
- The economic reason is not that entry never matters. Entry raises variety. The key point is that entry does *not* do much to the aggregate markup.
- Figure 5 explains why. When entry rises, all firms' markups fall directly, but small low-markup firms shrink by more than large high-markup firms.
- This compositional effect almost offsets the direct effect, leaving the aggregate markup nearly unchanged.
- So “more competition through entry” is too blunt an instrument for reducing markup distortions in a heterogeneous-firm economy.

5.6 Table 5, Table 6, and Table 7: oligopoly results

TABLE 5
PARAMETERIZATION, OLIGOPOLY

	DATA		CASES		
Calibration targets:					
Aggregate markup, $\mathcal{M}$	1.1–1.4	1.05	1.15	1.25	1.35
Top-4 sales share, CR4	.43	.37	.43	.43	.43
Top-20 sales share, CR20	.72	.76	.72	.72	.72
Materials share	.45	.45	.45	.45	.45
Regression coefficient, $\hat{b}$	-.21	-.21	-.21	-.21	-.21
Parameter values:					
Pareto tail, ξ		28.08	8.51	5.15	3.72
Elasticity of substitution within sectors, γ		59.69	12.76	7.16	5.21
Elasticity of substitution between sectors, η		1.62	1.35	1.15	.99
Average number of firms per sector, N		415	359	143	112
Weight on value-added, ϕ		.70	.58	.46	.30

NOTE.—Calibrated parameters for our oligopoly model. We calibrate the Pareto tail ξ , the within- and between-sector elasticities of substitution γ and η , the sunk entry cost κ , and the weight on value-added ϕ to match the targets shown. In practice, we choose the average number of firms N per sector and back out the sunk cost κ that rationalizes N . The cross-sectional regression is of the change over time in sector-level labor shares on the change in sector-level HHIs, as discussed in the text. All other parameters are assigned as in panel A of table 1.

TABLE 6
MARKUP DISPERSION AND PRODUCTIVITY LOSSES, OLIGOPOLY

	AGGREGATE MARKUP, $\mathcal{M}$							
	1.05		1.15		1.25		1.35	
	$\mu_i(s)$	$\mu_u(s)$	$\mu_i(s)$	$\mu_u(s)$	$\mu_i(s)$	$\mu_u(s)$	$\mu_i(s)$	$\mu_u(s)$
Cost-weighted distribution of markups:								
25th percentile markup	1.04	1.02	1.12	1.09	1.21	1.17	1.29	1.25
50th percentile markup	1.05	1.04	1.14	1.11	1.23	1.19	1.32	1.27
75th percentile markup	1.06	1.07	1.16	1.17	1.27	1.26	1.37	1.36
90th percentile markup	1.07	1.10	1.21	1.27	1.33	1.41	1.45	1.54
99th percentile markup	1.11	1.20	1.35	1.57	1.57	1.90	1.78	2.24
Aggregate-productivity losses (%):								
Gross output	2.99		3.19		3.32		3.81	
Value-added	5.55		6.85		8.89		12.52	
Value-added, $\mathcal{M} = 1$	5.45		6.02		6.51		7.74	

NOTE.—Cost-weighted steady-state distribution of firm-level markups $\mu_u(s)$ and sector-level markups $\mu_i(s)$ and the implied aggregate-productivity losses for our oligopoly model. Gross-output aggregate-productivity loss is $(Z - Z^*)/Z^* \times 100$, and similarly for the value-added aggregate-productivity loss. To isolate the effect of misallocation on value-added aggregate productivity, we also report the value-added aggregate-productivity loss with the same amount of markup dispersion but holding $\mathcal{M} = 1$, to eliminate the distortion between value-added and materials; see text for details.

Read:

	STEADY-STATE COMPARISONS (% change)						WELFARE (% change)
	Y	G	L	N	K	Z	
$M = 1.05$							
Efficient	20.1	15.9	4.8	-29.5	30.3	1.8	8.71
Uniform subsidy	14.2	9.4	6.0	3.7	23.1	.1	.58
Size-dependent subsidy	15.3	6.1	-1.1	-31.4	6.1	1.7	7.97
Entry subsidy	-2.0	-2.4	-1.1	-28.9	-2.5	-1.2	.50
$M = 1.15$							
Efficient	55.9	39.9	14.9	-10.5	94.0	1.9	14.66
Uniform subsidy	50.5	34.4	17.0	9.8	86.6	1.1	5.14
Size-dependent subsidy	4.1	4.4	-1.8	-18.4	4.5	.8	8.70
Entry subsidy	-5.1	-5.5	-1.0	-8.7	-2.7	-1.1	.12
$M = 1.25$							
Efficient	112.6	79.0	25.3	-1.7	206.6	3.1	26.76
Uniform subsidy	108.4	75.2	28.4	15.3	201.1	3.0	15.20
Size-dependent subsidy	3.0	3.0	-2.4	-14.5	3.1	.2	9.34
Entry subsidy	.9	1.0	.4	1.9	1.2	.4	.01
$M = 1.35$							
Efficient	204.0	142.7	35.3	3.6	412.1	5.0	48.63
Uniform subsidy	201.8	141.0	30.6	20.5	411.9	5.6	32.38
Size-dependent subsidy	2.8	2.7	-3.1	-12.7	2.7	-.1	11.28
Entry subsidy	8.3	9.4	3.4	11.1	11.2	5.2	.44

NOTE.—The first six columns report the percentage change from the initial distorted steady state with to the new steady state. The last column reports the consumption-equivalent welfare gains (including transitional dynamics). The alternative policies are (i) the efficient allocation, where all markups are removed; (ii) a uniform subsidy that eliminates the aggregate markup; (iii) size-dependent subsidies that eliminate misallocation and the entry distortion; and (iv) the optimal entry subsidy (or tax).

- Table 5 gives the oligopoly calibration. It targets CR4, CR20, the materials share, and the sector labor-share versus HHI slope.
- Table 6 shows that the oligopoly model generates much more markup dispersion than the benchmark monopolistic-competition model. At $M = 1.15$, the 99th percentile firm-level markup is about **1.57**, and the gross-output productivity loss from misallocation is about **3.19%**.
- Table 7 shows that the total welfare costs are also larger in oligopoly. At $M = 1.15$, the efficient allocation yields a welfare gain of about **14.66%**, versus **8.67%** in the benchmark model.
- Another useful contrast is on entry: in the benchmark model there are too few firms, whereas in the oligopoly model there can be too many firms in the distorted steady state. But quantitatively, the entry channel remains small relative to the aggregate markup and misallocation channels.

5.7 Appendix Figure A1 and Appendix Figure B1: how to read markup time series

Read:

- Appendix Figure A1 compares the cost-weighted and sales-weighted average markup in Computat. The sales-weighted average rises much more, illustrating the role of increasing dispersion.
- Appendix Figure B1 constructs the cost-weighted aggregate markup directly from Census data. Under constant returns to scale, the aggregate markup rises from roughly **1.20** in 1972 to a peak around **1.40** in 2002 before easing to about **1.33** in 2012.
- These appendix figures are important because they clarify what part of “rising markups” is an aggregate wedge and what part is dispersion.

6 Short summary

- **Method.** The paper builds a dynamic heterogeneous-firm model with endogenous markups, entry, and capital accumulation, and calibrates it to U.S. manufacturing moments.
- **Main conceptual result.** The aggregate markup relevant for macro wedges is the *cost-weighted average markup*. The sales-weighted arithmetic average also loads on markup dispersion and therefore overstates the aggregate wedge.

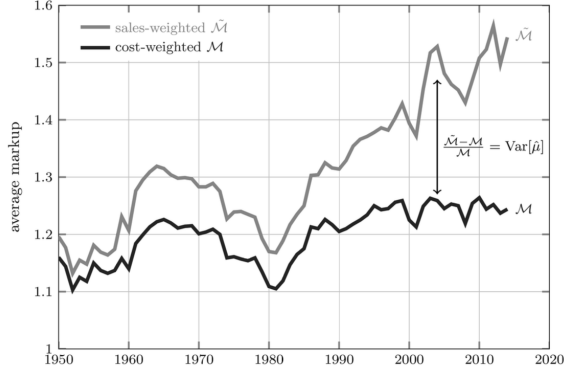


FIG. A1.—Cost-weighted versus sales-weighted average markups, Compustat: the sales-weighted average $\tilde{\mathcal{M}}$ of firm-level markups in Compustat data, as in De Loecker, Eeckhout, and Unger (2020), and the cost-weighted average of firm-level markups $\mathcal{M}$. The former is higher and has increased by a larger amount. The proportional difference between the two averages reflects the cross-sectional dispersion in markups, which has been increasing.

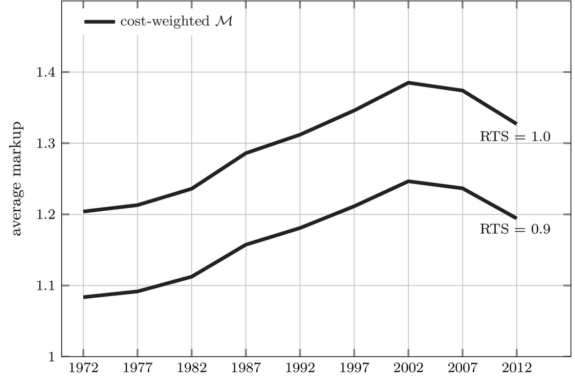


FIG. B1.—Aggregate markup from Census data: cost-weighted aggregate markup $\mathcal{M}$ computed from the firm-level markups $\mu_{i,t}(s)$ constructed with micro data from the US Census of Manufactures from 1972 to 2012, as discussed in the text, for different values of the returns to scale (RTS). Our benchmark model assumes constant RTS, $\text{RTS} = 1$, but our results are robust to lower RTS.

- **Main quantitative result.** Welfare costs of markups are large and convex in the level of the aggregate markup, with the aggregate-markup and misallocation channels accounting for most of the losses.
- **Policy lesson.** Uniform output subsidies that offset the aggregate wedge are powerful, size-dependent subsidies that correct misallocation are also valuable, but entry subsidies are a weak way to fight market power in a heterogeneous-firm economy.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the welfare cost of markups in a dynamic economy with heterogeneous firms can be very large.
- It also establishes that one cannot summarize these costs with a representative-firm statistic or with a sales-weighted arithmetic average markup.
- The economically relevant decomposition is into an aggregate-markup wedge, misallocation from markup dispersion, and a comparatively small entry distortion.

7.2 Economic intuition

- Markups hurt the economy in two different ways. First, they reduce the overall scale of production, investment, and employment, like a common output tax. Second, when high-markup firms are also the most productive firms, they are inefficiently small relative to the planner's allocation.
- Entry is much less powerful than one might think because heterogeneous firms respond asymmetrically to extra competition: small low-markup firms contract more than large high-markup firms.

- This is why raising competition through entry barely moves the aggregate markup even if it moves firm-level markups.

7.3 Takeaway

This paper is important because it links the modern markup literature to quantitative welfare analysis in a disciplined dynamic setting. Its central lesson is that rising markups should be interpreted through the lens of both levels and dispersion. If one wants to think about policy, the first-order object is the aggregate wedge created by the cost-weighted markup, while the second-order but still important object is the misallocation created by markup dispersion. In the paper's calibrations, both matter a lot, while the entry margin matters surprisingly little.